

Sai Phani Krishna Arumalla

AI/ML Engineer | LangGraph · RAG · PyTorch · AWS

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Summary

MS Computer Science candidate at the University of South Florida (GPA: 3.66, May 2026) specializing in production LLM systems, multi-agent orchestration, RAG pipelines, and full-stack AI application development. Builder of three deployed AI systems: DealMind, ArchGuard, and OmniScan Engine, each live and used by real users. Passionate about shipping AI that solves real problems, not just demos.

Education

Master of Science, Computer Science

Aug 2024 – May 2026

University of South Florida

- Machine Learning
- Deep Learning
- Natural Language Processing
- Distributed Systems
- Applied Artificial Intelligence

Bachelor of Technology, Information Technology

Dec 2020 – Apr 2024

CVR College of Engineering

- Data Structures and Algorithms
- Database Management Systems
- Software Engineering
- Computer Networks
- Machine Learning Fundamentals

Professional Experience

Sole Developer, ML Engineer

Jun 2026 – Jul 2026

DomainTune (Independent Project)

- Ran SFT with QLoRA $r=16$ + DoRA on Qwen2.5-3B-Instruct for 200 steps on a free Colab T4, then DPO alignment over 50 steps on the SFT model's own validation-set failures ($\beta=0.1$)
- Discovered and fixed the dominant error source: base model ignored 'amounts in millions' qualifiers in 10-K filings, outputting 60,922 instead of 60,922,000,000. Injecting a [UNIT NOTE] into every training example lifted revenue accuracy from 24% to 86%
- DPO alignment lifted all-five-fields-correct rate from 2% to 20% (10x). EPS within \$0.01: 88% to 98%. Validated with bootstrap CI ($n=2000$), ROUGE-1, and LLM-as-judge; 95% CI non-overlapping on all numeric fields
- Published fine-tuned model to HuggingFace ([saiphanikrishna/domaintune-qwen2.5-3b-sec](#)) and built React demo showing Base vs DPO side-by-side on live tickers

Sole Developer, AI/ML Engineer

Feb 2026 – Mar 2026

DealMind (Independent Project)

- Designed multi-agent orchestration using CrewAI and LangGraph with specialized agents for research, analysis, and synthesis, replacing hours of manual financial due diligence
- Built production RAG pipeline using ChromaDB vector database with document ingestion, chunking strategy optimization, and retrieval tuning for grounded LLM outputs
- Implemented dynamic model routing scoring task complexity to route between Claude and Gemini APIs, reducing inference cost while maintaining output quality
- Built HITL oversight checkpoints flagging low-confidence outputs for human review before finalization
- Instrumented W&B Weave observability tracking hallucination rate, latency, and cost-per-request in production
- Deployed full-stack application: Python/FastAPI backend, React/SSE frontend, Docker, AWS (EC2, S3), CI/CD via GitHub Actions

Sole Developer, AI/ML Engineer

Mar 2026 – Apr 2026

ArchGuard (Independent Project)

- Built LangGraph stateful agent that parses codebases via Tree-sitter AST analysis, embeds structural representations in ChromaDB, and reasons over retrieved context to detect compliance violations
- Designed Golden Dataset evaluation benchmark from scratch, implementing LLM-as-judge grading and validating production performance at 0.92 F1-Score and 1.00 Recall
- Built and deployed MCP (Model Context Protocol) server enabling standardized agent-to-tool integration across enterprise developer toolchains
- Distributed as PyPI package, GitHub Action, and MCP server, enabling adoption without workflow changes
- Applied HITL escalation for low-confidence compliance judgments, ensuring high-stakes decisions route to human review

Accessibility Services Student Assistant

Dec 2025 – May 2026

University of South Florida

- Preprocessed 100+ academic documents into structured, machine-readable formats: tagged PDFs, caption files, and semantic annotations, applying document remediation techniques foundational to AI retrieval and assistive NLP systems serving students with visual and cognitive disabilities.
- Translated disability profiles into assistive technology configurations for 50+ students per semester, building hands-on experience in human-centered AI deployment and ADA/Section 508 compliance standards increasingly mandated in enterprise AI products.
- Coordinated adaptive exam environments for students with motor, sensory, and cognitive impairments, ensuring seamless integration of assistive input devices and accessibility software.

Sole Developer, ML Engineer

Jan 2026 – Feb 2026

OmniScan Engine (Independent Project)

- Trained Vision Transformer (ViT) and SE-ResNet ensemble using PyTorch on multimodal inputs (video, audio, image) for deepfake classification
- Implemented XAI confidence heatmaps for model explainability, surfacing uncertainty visualizations to end users
- Built FastAPI inference API and React frontend with automated PDF audit report generation
- Deployed on AWS (EC2, S3) with Docker containerization and CI/CD automation via GitHub Actions

Technical Skills

Agentic AI & Orchestration: LangGraph, CrewAI, MCP (Model Context Protocol), Multi-Agent Orchestration, Tool Use, HITL Checkpoints, Dynamic Model Routing, Agentic Workflow Design

LLM & Generative AI: Anthropic Claude API, Gemini API, OpenAI-compatible APIs, Prompt Engineering, RAG, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Fine-Tuning Concepts (LoRA/QLoRA), Hugging Face Transformers

Machine Learning & Deep Learning: PyTorch, Scikit-learn, TensorFlow (familiar), Supervised Learning, Unsupervised Learning, Computer Vision, Vision Transformer (ViT), SE-ResNet, Model Evaluation (F1/Precision/Recall/ROC-AUC), Experiment Tracking (W&B Weave)

RAG & Vector Databases: Retrieval-Augmented Generation, ChromaDB, FAISS (familiar), Pinecone (familiar), Embeddings, Semantic Search, Document Ingestion, Chunking Strategies, Retrieval Tuning

Programming Languages: Python, JavaScript, Java, TypeScript (familiar), SQL, Bash

Backend & APIs: FastAPI, REST API Design, Server-Sent Events (SSE), Pydantic, PostgreSQL, MongoDB, ChromaDB

Infrastructure & MLOps: AWS (EC2, S3), Docker, CI/CD (GitHub Actions), Linux, Git, W&B Weave, PyPI, Vercel

Frontend:

React, JavaScript, Vite, Tailwind CSS, Shadcn UI, Real-Time SSE Streaming